

TIC TOC TOE:

```
board = ['__'] * 9

# All winning combinations (rows, columns, diagonals)
wins = [
    [0, 1, 2], [3, 4, 5], [6, 7, 8],
    [0, 3, 6], [1, 4, 7], [2, 5, 8],
    [0, 4, 8], [2, 4, 6]
]

def print_board():
    print()
    print(board[0], board[1], board[2])
    print(board[3], board[4], board[5])
    print(board[6], board[7], board[8])
    print()

def check_win(player):
    for combo in wins:
        if board[combo[0]] == board[combo[1]] == board[combo[2]] ==
player:
            return True
    return False

player = 'X'
print("Positions are numbered 0 to 8, left to right, top to bottom.")
print_board()

for turn in range(9):
    pos = int(input(f"Player {player}, enter position (0-8): "))

    if board[pos] != '__':
        print("That position is already taken. Try again.")
        continue
```

```
board[pos] = player
print_board()

if check_win(player):
    print(f"Player {player} wins!")
    break

if '___' not in board:
    print("It's a draw!")
    break

# switch player
if player == 'X':
    player = 'O'
else:
    player = 'X'
7
8
```

OUTPUT:

20s



... Positions are numbered 0 to 8, left to right, top to bottom

```
__ __ __  
__ __ __  
__ __ __
```

Player X, enter position (0-8): 0

```
X __ __  
__ __ __  
__ __ __
```

Player O, enter position (0-8): 4

```
X __ __  
__ O __  
__ __ __
```

Player X, enter position (0-8): 2

```
X __ X  
__ O __  
__ __ __
```

Player O, enter position (0-8): 6

```
X __ X  
__ O __  
O __ __
```

Player X, enter position (0-8): 1

```
X X X  
__ O __  
O __ __
```

Player X wins!

8